

Natural language and external conventions: re-examining *per**

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Abstract

We argue that *per*-phrases like *two grams per liter* exhibit a dual interpretive pattern. When such phrases saturate predicates of quantities in simplex dimensions, like weight or distance, they are interpreted compositionally within grammar; by contrast, when referring to quantities in quotient dimensions, like density or speed, they verbalize technical notation using a type of *math speak*. Akin to mixed quotations, such verbalizations get their meanings from extra-grammatical conventions. Supporting our analysis, *per*-phrases that refer to quotient dimensions, unlike their simplex counterparts, allow meaning-preserving substitution with expressions that unambiguously verbalize technical notation. Furthermore, these same expressions are opaque to syntactic operations. Our arguments challenge the view that *per* encodes quantity division compositionally, and underscore the distinction between meanings built in grammar and those licensed by extra-grammatical conventions.

Keywords: Measurement, Scales, Quantity Calculus, Quotient Quantities, Mixed Quotation

*For thought-provoking questions and suggestions, we thank three anonymous reviewers and the editor Luka Crnić. For stimulating discussion during earlier stages of the research reported here, we also thank audiences at *SALT 32* and the workshop *Degrees and questions at the interfaces* at The Hebrew University of Jerusalem, especially Liz Coppock, Luka Crnić, Jenny Doetjes, Yosef Grodzinsky, Marcin Morzycki, and Jessica Rett. This research was supported in part by funding from the Social Sciences and Humanities Research Council, through SSHRC Insight grants 435-2023-0699 and 435-2023-0218.

1 Introduction

Measurements relate entities to *quantities* along specific *dimensions*, such as weight (WT) or volume (VOL). For instance, a portion of mercury might weigh 130g and have a volume of 10mL. The sciences have developed a formal system, often called a *quantity calculus*, to handle and manipulate such measurements in theoretical models. A central feature of this system is its capacity to capture *complex dimensions* and *complex quantities*, such as *quotient dimensions* formed through quantity division. For example, the expression $13 \frac{\text{g}}{\text{mL}}$ denotes a measure of density in the quotient dimension $\frac{\text{WT}}{\text{VOL}}$.¹ In this paper, we investigate whether natural language can combine phrases referring to quantities like weight and volume to derive quotient quantities in a manner akin to the quantity calculus. Thus, we ask whether semantic composition in language can mirror the division-like operations used in technical notation.

Coppock (2021, 2022) argues that natural language aligns with the quantity calculus in this manner. Specifically, Coppock proposes that the preposition *per* can denote a division operation over quantities, thereby establishing a direct correspondence between linguistic expressions like *thirteen grams per milliliter* and the symbolic representations in the quantity calculus, such as $13 \frac{\text{g}}{\text{mL}}$. Coppock formalizes this interpretation with the lexical entry in (1), where r and q range over quantities.

$$(1) \quad \llbracket \text{per} \rrbracket = \lambda r. \lambda q. \frac{q}{r}$$

This proposal enables a technically straightforward compositional analysis for sentences where a measure phrase containing *per* invokes a complex dimension such as density. The truth of (2), for instance, seems to require the denotation of *thirteen grams per milliliter* to fall within the extension of the predicate *density of that sample of mercury*. If *per* could form quantities like $13 \frac{\text{g}}{\text{mL}}$, as enabled by (1), it would naturally supply a denotation of the requisite sort. As this would appear to capture speaker intuitions, data like (2) seem to support the idea that quantity division is a possible semantic operation within natural language composition.

- (2) The density of that sample of mercury is thirteen grams per milliliter.

However, we argue here that this seemingly straightforward take on the data does not withstand closer scrutiny. Specifically, we challenge the assumption that *per* can be associated with the lexical entry in (1). Furthermore, we argue against any compositional interpretation that associates *per*-phrases with either quotient quantities or quantity division. Such an association is neither required to explain data like (2), nor sufficient to account for the broader compositional behaviour of *per*-phrases.

In the remainder of this introductory section, we preview these arguments. To begin, consider the sentence in (3), focusing on the sequence *four times ten to the thirteenth*. It is clear that this sequence refers to a specific number—the one expressed in technical notation as 4×10^{13} . But, how is this referent determined?

- (3) The distance from Earth to Proxima Centauri is four times ten to the thirteenth kilometers.

¹Here and elsewhere, we deliberately use the more intuitive term “weight” rather than the more technically correct term “mass.”

It seems unlikely that natural language compositionally encodes operations like exponentiation through phrases such as *to the thirteenth*. Instead, or so we will argue, this sequence is a case of *math speak*. By this we mean that the sequence functions as a way of verbalizing an expression in technical notation, here ' 4×10^{13} '. Critically, the question of how the mathematical term ' 4×10^{13} ' refers to a specific number belongs to the domain of technical notation, not to the grammar of English. Thus, plausibly, the denotation of the phrase *four times ten to the thirteenth* is not derived from semantic composition based on the syntactic structure of English, but instead relies on conventions external to natural language grammar.

With the quantity calculus serving as the technical notation, we suggest that this possible interpretation of (3) extends naturally to (2). Specifically, we propose that the sequence *grams per milliliter* in (2) can be math speak as well, functioning as a verbalization of the quantity calculus term ' $\frac{g}{mL}$ '. Its denotation, $\frac{g}{mL}$, arises not from compositional semantic operations within the grammar of English, but rather from its alignment with the conventional use of the quantity calculus term. If this account is correct, then the data in (2) do not provide evidence for the lexical entry for *per* in (1). Instead, the connection between the sequence *thirteen grams per milliliter* and its referent depends on external conventions rather than internal composition.

This is not to suggest that all uses of *per* are interpreted in this way. On the contrary, we agree with Coppock (2021, 2022) and Bale and Schwarz (2022) that the denotation of a *per*-phrase can also arise through ordinary semantic composition. Evidence for this lies in the fact that *per*-phrases can saturate predicates that do not apply to the kinds of quantities that would arise from quantity division. The sentence in (4) exemplifies this point.

(4) This sample of mercury weighs thirteen grams per milliliter.

The predicate *weigh* relates entities to weights, not densities. Consequently, the phrase *thirteen grams per milliliter* in (4) cannot denote $13 \frac{g}{mL}$, a quantity within the dimension $\frac{WT}{VOL}$. For this case, then, the entry in (1) would not have the right effect to begin with. Interpreting (4) seems to require a lexical entry for *per* that is different from (1).

Assuming that *per* is ambiguous, Coppock (2022) proposes such an entry, in addition to (1). While this second entry does not yield a quotient quantity, it still incorporates quotient quantities and the division line within the description of its denotation. In contrast, Bale and Schwarz (2022) argue that, for sentences like (4), appealing to quotient quantities can be avoided altogether. Their entry for *per* employs a dividing line, but merely to derive a pure number, not a quotient quantity. Expanding on this, we will show here that neither Coppock's interpretation nor that of Bale and Schwarz necessitates quotient quantities or dividing lines. Both analyses can be reformulated exclusively in terms of multiplication.

We believe that two main insights emerge from our discussion. First, certain *per*-phrases require interpretations that are outside the purview of natural language grammar. Why might this be so? Given how common these expressions are, why wouldn't grammar enable composition of the relevant meanings? We suspect that the

concept of quantity division, a fairly recent innovation in mathematics, lacks a counterpart in the inventory of concepts accessible to natural language semantics. Thus, we advance the *No Division Hypothesis* in (5).

- (5) **No Division Hypothesis:** Quantity division is not available as an operation for semantic composition.

Second, our findings highlight the need for caution when interpreting phrases with mathematical or technical correlates. Given the existence of math speak, it is possible that they owe their meanings to extra-grammatical mechanisms. Thus, to establish an analysis in terms of semantic composition within grammar, it is necessary to show that the meanings derived are not instead given by external conventions.²

The remainder of this paper proceeds as follows. In Section 2, we examine alternative interpretations of *per*, highlighting analyses that avoid reliance on quotient quantities and division operations. We show that these analyses can be reformulated using pure numbers and multiplication, offering a simpler and more intuitive account of the semantics *per*. Section 3 investigates cases where *per*-phrases are interpreted non-compositionally, being cases of math speak. By drawing parallels with the phenomenon of mixed quotation, we argue that these interpretations reflect the interplay between linguistic grammar and external systems of representation. Finally, in Section 4, we conclude by reviewing the implications of our findings.

2 Pure numbers and the interpretation of *per*

The interpretation of *per* raises intriguing questions about whether its semantics inherently relies on the creation of quotient quantities. While it seems to evoke division in phrases like *thirteen grams per milliliter* or *thirty miles per hour*, a closer inspection reveals that alternative explanations can suffice for relevant data. Exploring these alternatives sheds light on how *per* contributes to the composition of meaning.

As noted, *per* can be used to refer to quantities in quotient dimensions such as density (weight divided by volume) or speed (distance divided by time), as in (6). However, as discussed in Bale and Schwarz (2022) and Coppock (2022), *per* can also appear in phrases that describe quantities in simplex dimensions, like weight or distance. To see this, consider the measure predicates *weigh* and *cover*. As (7) shows, these verbs take complements like *thirteen grams* or *thirty miles*—that is, they apply to quantities in the dimensions weight and distance, respectively. Crucially, however, as illustrated in (8), *per*-phrases can occupy the same complement positions. Hence, in such cases, *per*-phrases describe quantities in these simplex dimensions.³

²Aligned with our proposal, Bylinina and Nouwen (2020) entertain the possibility that a sentence like *Twelve plus twelve is twenty four* is in some sense not part of English, but a form of “mathematical English”, a separate linguistic system whose lexicon overlaps with the lexicon of English. Similarly, Morzycki (2017) introduces the concept of a “semantic virus”, a broad term encompassing the hypothesis that certain quasi-mathematical terms may lack an interpretation within the narrow linguistic grammar. Morzycki suggests that such terms might be interpreted extra-linguistically. The sort of math speak we appeal to here could be considered a type of semantic virus in this sense.

³Bale and Schwarz (2022) point out that a sentence like *This cube weighs what that cube weighs* cannot be understood as a statement about the two cubes’ densities, as expected if *weigh* is a predicate of weights, not measures of density. That said, as pointed out by two reviewers, it seems that under certain conditions, sentences with *weigh* can be understood as though they contained a covert *per*-phrase. For example, this is the case for sentences like *Water weighs more than oil*, which plausibly conveys that per unit of volume,

- (6) a. The density of that sample of liquid is thirteen grams per milliliter.
b. The speed of that train is thirty miles per hour.
- (7) a. This sample of liquid weighs thirteen grams.
b. During its six-hour trip, the train (only) covered thirty miles due to the frequent stops it had to make.
- (8) a. This sample of liquid weighs thirteen grams per milliliter.
b. During its six-hour trip, the train (only) covered thirty miles per hour due to the frequent stops it had to make.

Both Coppock (2022) and Bale and Schwarz (2022) argue that the denotation provided in (1), repeated below, cannot adequately account for these types of sentences. In particular, a quotient quantity like $13 \frac{\text{g}}{\text{ml}}$ cannot represent a unit of weight, nor can $30 \frac{\text{mi}}{\text{hr}}$ (= 30 mph) represent a unit of distance.

$$(1) \quad \llbracket \text{per} \rrbracket = \lambda r. \lambda q. \frac{q}{r}$$

To address this limitation, both papers propose alternative interpretations of *per*. In what follows, we review these proposals. We will see that they can be reframed to eliminate not only a reliance on quotient quantities, but also the necessity of any type of division operation.

Coppock (2022) suggests that, in addition to permitting the denotation in (1), *per* may also be interpreted via the entry in (9). Here q and r are variables over quantities, type d , f ranges over possible (denotations of) measure predicates, type $\langle d, \langle e, t \rangle \rangle$, and x ranges over individuals, type e .

$$(9) \quad \llbracket \text{per} \rrbracket = \lambda q. \lambda r. \lambda f. \lambda x. \frac{\max(\{d: f(d)(x)\})}{\mu_{\text{dim}(q)}(x)} = \frac{r}{q}$$

The function in (9) can be broken down into two essentially components. First it takes two quantities as arguments (q and r) to create the quotient quantity $\frac{r}{q}$. Second, it takes a measure predicate and a subject (referent) as arguments (f and x) to create another quotient quantity. The numerator of this second quotient quantity, $\max(\{d: f(d)(x)\})$, denotes the maximal measure of the subject x as determined by the measure predicate f . The denominator $\mu_{\text{dim}(q)}(x)$ denotes the measure of the subject via the canonical measure function associated with the dimension of the quantity q . The final truth conditions depend on a comparison of these two quotient quantities, mapping to true if and only if they are equivalent.

To develop an intuitive understanding of this function, it helps to examine concrete examples. Consider the sentence in (8a). Under Coppock’s analysis, $\llbracket \text{milliliter} \rrbracket$ (i.e., mL) and $\llbracket \text{thirteen grams} \rrbracket$ (i.e., 13g) represent the first and second quantity arguments respectively, whereas $\llbracket \text{weighs} \rrbracket$ (i.e., $\lambda d. \lambda x. \mu_{\text{WT}}(x) = d$) and $\llbracket \text{this sample of liquid} \rrbracket$ instantiate the measure predicate and subject arguments. Using ‘sample’ as shorthand for the subject referent, the denotation in (9) derives the truth conditions in (10).

$$(10) \quad \frac{\max(\{d: \mu_{\text{WT}}(\text{sample}) = d\})}{\mu_{\text{dim}(\text{mL})}(\text{sample})} = \frac{13\text{g}}{\text{mL}}$$

water has a greater weight than oil. We leave it to future work to investigate the conditions that enable such readings.

If we make the reasonable assumption that the canonical measurement associated with milliliters is volume (e.g. $\mu_{\text{dim}(\text{mL})} = \mu_{\text{VOL}}$), (10) ends up being equivalent to (11)—a formula that is true if and only if the ratio of the sample’s weight to its volume is equivalent to the ratio of 13 grams to 1 milliliter.

$$(11) \quad \frac{\mu_{\text{WT}}(\text{sample})}{\mu_{\text{VOL}}(\text{sample})} = \frac{13\text{g}}{\text{mL}}$$

A similar analysis can be given for (8b). For example, suppose that $\llbracket \text{hour} \rrbracket = \text{hr}$, $\llbracket \text{thirty miles} \rrbracket = 30\text{mi}$, and $\llbracket \text{covered} \rrbracket = \lambda d. \lambda x. \mu_{\text{DIST_TRAVELLED}}(x) = d$, where $\mu_{\text{DIST_TRAVELLED}}$ measures the distance travelled by an entity. The function in (9) would then assign *the train covered thirty miles per hour* the truth conditions in (12).

$$(12) \quad \frac{\max(\{d: \mu_{\text{DIST_TRAVELLED}}(\text{train}) = d\})}{\mu_{\text{dim}(\text{hr})}(\text{train})} = \frac{30\text{mi}}{\text{hr}}$$

If we further assume that $\mu_{\text{dim}(\text{hr})}(\text{train})$ measures the time travelled by the train (that is, $\mu_{\text{dim}(\text{hr})} = \mu_{\text{TIME_TRAVELLED}}$), we would get the final truth conditions in (13). Thus, the sentence winds up true if and only if the ratio of the distance travelled by the train to the time it travelled is equal to the ratio of thirty miles to one hour.

$$(13) \quad \frac{\mu_{\text{DIST_TRAVELLED}}(\text{train})}{\mu_{\text{TIME_TRAVELLED}}(\text{train})} = \frac{30\text{mi}}{\text{hr}}$$

On cursory inspection, Coppock’s analysis of *per* in these types of sentences might seem to rely on quotient quantities. Specifically, it appears to suggest that the function associated with this alternate interpretation creates two quotient quantities rather than one. However, this impression is actually an illusion that is created by the notation. Given basic algebraic principles, the truth conditions in (11) and (13) can be restated equivalently as in (14). Accordingly, the entry for *per* in (9) could be restated equivalently as in (15).

$$(14) \quad \text{a. } \mu_{\text{WT}}(\text{sample}) = \frac{\mu_{\text{VOL}}(\text{sample})}{\text{mL}} * 13\text{g}$$

$$\text{b. } \mu_{\text{DIST_TRAVELLED}}(\text{train}) = \frac{\mu_{\text{TIME_TRAVELLED}}(\text{train})}{\text{hr}} * 30\text{mi}$$

$$(15) \quad \llbracket \text{per} \rrbracket = \lambda q. \lambda r. \lambda f. \lambda x. \max(\{d: f(d)(x)\}) = \frac{\mu_{\text{dim}(q)}(x)}{q} * r$$

In (14), unlike in (11) and (13), there is no term describing a quotient quantity representing a measure of density or speed. Instead, the dividing line in (14), as seen in the terms $\frac{\mu_{\text{VOL}}(\text{sample})}{\text{mL}}$ and $\frac{\mu_{\text{TIME_TRAVELLED}}(\text{train})}{\text{hr}}$, serves only to compute a pure number. For instance, if the sample’s volume were 20mL, then $\frac{\mu_{\text{VOL}}(\text{sample})}{\text{mL}}$ would simplify to $\frac{20\text{mL}}{\text{mL}}$, which equals 20. Similarly, if the time travelled by the train were 6hr, then $\frac{\mu_{\text{TIME_TRAVELLED}}(\text{train})}{\text{hr}}$ would be equivalent to $\frac{6\text{hr}}{\text{hr}}$, which reduces to 6. This property is not specific to these examples. The value of $\frac{\mu_{\text{dim}(q)}(x)}{q}$ will always yield a pure number, as the unit in the numerator cancels with the unit in the denominator.⁴

⁴To capture the intuited meaning of cases like *The cost per ton is \$100*, Coppock (2022) proposes a further lexical entry for *per*. However, again consistent with the No Division Hypothesis, this lexical entry also does not invoke division of quantities in different dimension.

Bale and Schwarz (2022) propose an interpretation of *per* that is conceptually similar to Coppock’s but differs in certain details, particularly from the version of Coppock’s entry given in (15). They argue that *per*-phrases like *thirteen grams per milliliter* denote simple quantities. This is derived by using one quantity (e.g., $\llbracket \text{milliliter} \rrbracket$ in the example above) to derive a pure number and treating the other (e.g., $\llbracket 13 \text{ grams} \rrbracket$) as a base unit. They suggest that *per* has an interpretation akin to the one in (16).⁵

$$(16) \quad \llbracket \text{per} \rrbracket = \lambda q. \lambda x. \lambda r. \frac{\mu_{\text{dim}(q)}(x)}{q} * r$$

Unlike Coppock (2022), Bale and Schwarz do not posit that *per* takes a measure predicate as an argument. Instead, they propose that *per* interacts with a covert pronoun (i.e., *pro*), whose value is determined anaphorically by other elements in the sentence. For instance, under their analysis, *thirteen grams per milliliter* and *30 miles per hour* would have the syntactic structure shown in (17).

- (17) a. $[_{MP} [_{MP} \text{thirteen grams}] [_{PP} \text{pro} [_{\text{per}} \text{milliliter}]]]$
 b. $[_{MP} [_{MP} \text{thirty miles}] [_{PP} \text{pro} [_{\text{per}} \text{hour}]]]$

In the context of the sentence in (8a), the covert pronoun *pro* is anaphorically linked to the subject, that is, *this sample of liquid*. As a result, the measure phrase *thirteen grams per milliliter* is interpreted as in (18).

$$(18) \quad \llbracket [_{MP} [_{MP} \text{thirteen grams}] [_{PP} \text{pro} [_{\text{per}} \text{milliliter}]]] \rrbracket = \frac{\mu_{\text{vol}}(\llbracket \text{pro} \rrbracket)}{\text{mL}} * 13\text{g} \\ = \frac{\mu_{\text{vol}}(\text{sample})}{\text{mL}} * 13\text{g}$$

Suppose that in the sentence in (8b), the covert pronoun *pro* is anaphorically linked to the phrase *its six-hour trip*, whose denotation we indicate as ‘train-trip’. For the measure phrase in (8b), this yields the interpretation in (19).

$$(19) \quad \llbracket [_{MP} [_{MP} \text{thirty miles}] [_{PP} \text{pro} [_{\text{per}} \text{hour}]]] \rrbracket = \frac{\mu_{\text{time}}(\llbracket \text{pro} \rrbracket)}{\text{hr}} * 30\text{mi} \\ = \frac{\mu_{\text{time}}(\text{train-trip})}{\text{hr}} * 30\text{mi}$$

Assuming equivalence of $\mu_{\text{TIME_TRAVELLED}}(\text{train})$ and $\mu_{\text{TIME}}(\text{train-trip})$, these denotations for the *per*-phrases derive the same truth conditions for the sentences in (8) as those derived under Coppock’s account, stated in (14) above.⁶

⁵The interpretation in (16) differs slightly from the one in Bale and Schwarz 2022. In Bale and Schwarz 2022, syntactic composition is interpreted as multiplication, eliminating the need for *per* to take the *r* quantity as its direct argument.

⁶The primary difference between Bale and Schwarz’s analysis on the one hand and Coppock’s on the other lies in the compositional structure. Bale and Schwarz hypothesize that the measure phrase *thirteen grams per milliliter* denotes a quantity of weight (in grams). The measure predicate then takes this quantity as an argument, asserting that the subject has a weight equivalent to this quantity. As a result, the composition of *This sample of liquid weighs thirteen grams* and *This sample of liquid weighs thirteen grams per milliliter* follows a parallel structure. In contrast, Coppock argues for a fundamentally different composition, where *per* takes the measure predicate as an argument rather than the measure predicate taking the measure phrase. It is worth noting that, while the two proposals are on a par for the purposes of the arguments in this paper, they might be dissociated by further data. For example, Coppock’s composition may not extend without further assumptions to cases like *The weight of this sample is thirteen grams per milliliter*, where the syntactic position of *this sample* seems to prevent it from serving as an argument of *per*. In contrast, no challenge arises from such cases under Bale and Schwarz’s analysis, since it seems that again nothing would prevent *pro* from co-referring with *this sample*.

Under these analyses of *per*, not only is reference to quotient quantities not required, but the dividing line can be eliminated altogether. For example, the number derived from $\frac{\mu_{\text{VOL}}(\text{sample})}{\text{mL}}$ is equivalent to the number computed by $\iota\{n: n * \text{mL} = \mu_{\text{VOL}}(\text{sample})\}$. Here we take ι to map a singleton set to its unique member, so $\iota\{n: n * \text{mL} = \mu_{\text{VOL}}(\text{sample})\}$ is the unique number n such that n times mL is equivalent to the measure of the sample's volume. Likewise, $\frac{\mu_{\text{TIME_TRAVELLED}}(\text{train})}{\text{hr}}$ and $\frac{\mu_{\text{TIME}}(\text{train-trip})}{\text{hr}}$ are equivalent to $\iota\{n: n * \text{hr} = \mu_{\text{TIME}}(\text{train-trip})\}$ and $\iota\{n: n * \text{hr} = \mu_{\text{TIME_TRAVELLED}}(\text{train})\}$, respectively. Thus, the function in (20) aligns with Coppock's interpretation of *per* in (9), while the function in (21) corresponds to Bale and Schwarz's interpretation in (16). The functions in (20) and (21) rely solely on multiplication in specifying their denotations. These restatements make it particularly obvious that for the proposed denotations for *per*, no quotient quantities needs to be computed.

$$(20) \quad \llbracket \text{per} \rrbracket = \lambda q. \lambda r. \lambda f. \lambda x. \max(\{d: f(d)(x)\}) = \iota\{n: n * q = \mu_{\text{dim}(q)}(x)\} * r$$

$$(21) \quad \llbracket \text{per} \rrbracket = \lambda q. \lambda x. \lambda r. \iota\{n: n * q = \mu_{\text{dim}(q)}(x)\} * r.$$

In summary, consistent with the No Division Hypothesis, neither Coppock's nor Bale and Schwarz's analyses require division to interpret *per*. A dividing line might be convenient to state the relevant denotation of *per*. However, both accounts can be re-imagined using pure numbers and multiplication only. Thus, these accounts do not entail a commitment to quantity division as an operation available in composition.

3 Non-compositional interpretations

We have so far demonstrated that phrases like *thirteen grams per milliliter* and *thirty miles per hour* allow for compositional interpretations that yield referents in simplex dimensions like weight and distance. We now turn to cases where these phrases refer to quantities in quotient dimensions such as density or speed. The evidence suggests that in such cases, meanings arise not from semantic composition within natural language, but from conventions governed by extra-grammatical symbolic systems. We suggest that these *per*-phrases can be unified with familiar cases of so-called mixed quotation, where natural language meaning is likewise shaped by external conventions.

3.1 Idioms, math speak, and external conventions

Let us begin with an examination of the phrase *pounds per square inch*, as used in (22). In many English-speaking countries (e.g., Canada), this phrase conventionally refers to a unit of tire pressure. Its behaviour, as we will see, suggests that not all *per*-phrases derive their meanings through ordinary semantic composition.

$$(22) \quad \text{The air pressure in this tire is 33 pounds per square inch.}$$

On the surface, this measure expression doesn't look any different from the *per*-phrases analyzed in the previous section. However, it has certain idiosyncratic properties that set it apart from garden variety cases like *grams per milliliter* and *miles per hour*. Even speakers who are accustomed to checking tire pressure at a gas station may not actually be familiar with *pounds per square inch*. Instead, they may only know

the phrase’s initialism *psi*, pronounced “pee ess aye”, as in (23), since it is just this initialism that typically appears written on the displays of pressure gauges and the sidewalls of tires. Thus, those speakers will treat the initialism as a unit noun, without connecting it to, or even being aware of, the full phrase that that it abbreviates.

- (23) The air pressure in this tire is 33 psi.

Given this, let us now imagine that such a speaker has just learned that *psi* serves as a shorthand for *pounds per square inch*—as was the experience of one of the authors. It seems intuitively clear that such a speaker is able to acquire the meaning of *pounds per square inch* by simply considering it a synonym of *psi*, without necessarily being cognizant of how a pound-force relates to a square inch—indeed without even being aware of what a pound-force is. Hence, although *pounds per square inch* appears to be syntactically complex, such a speaker would not arrive at its meaning through semantic composition. The relation between *psi* and *pounds per square inch* in this scenario is then not unlike the relation between an idiomatic phrase like *kick the bucket* and a monomorphemic synonym like *die*. In the speaker’s grammar, *pounds per square inch* is, essentially, an idiom.

From this idiomatic case, a broader lesson emerges. Just as *pounds per square inch* functions as an unanalyzed unit, so too can *per*-phrases that describe other quotient dimensions. Their meanings, we argue, might likewise bypass ordinary semantic composition. Let’s reconsider the phrases *grams per milliliter* and *miles per hour* discussed in the previous section, focussing on cases where such phrases describe quantities in quotient dimensions like density or speed, as in the examples in (6), repeated below.

- (6) a. The density of that sample of liquid is thirteen grams per milliliter.
b. The speed of that train is thirty miles per hour.

We acknowledge that unlike *pounds per square inch* in the scenario considered above, the *per*-phrases in (6) are *not* idioms akin to phrases like *kick the bucket*. Even in data like (6), such *per*-phrases are productive. That is, sequences that could substitute for *grams per milliliter* and *miles per hour* in (6) can be generated by replacing either of the two unit nouns, as in, say, *ounces per gallon* or *inches per second*.

However, this productivity does not force the conclusion that the *per*-phrases in (6) get their meanings through semantic composition within the grammar of English. In fact, there is evidence to the contrary. To put this evidence in the proper light, let us step back and consider again the sequence *four times ten to the thirteenth*, as in sentence (3), repeated here from Section 1.

- (3) The distance from Earth to Proxima Centauri is four times ten to the thirteenth kilometers.

Sequences of this sort are again productive. The numerals it contains, and also the multiplier *times*, can be replaced to obtain new expressions with the same external distribution, such as *six times ten to the fifteenth* or *six divided by ten to the thirteenth*. And yet, as previewed in Section 1, it would seem unreasonable to assume that *four times ten to the thirteenth* receives a compositional interpretation entirely within the

grammar of English. In particular, it seems unlikely that for the sub-sequence *ten to the thirteenth*, there are independently motivated lexical entries and principles of composition that reference the operation of exponentiation to deliver the number 10^{13} as the intended referent.

We suggest that it is more plausible that the sequence *ten to the thirteenth* is a conventional way of *verbalizing* the mathematical expression ‘ 10^{13} ,’ hence constitutes a form of *math speak*. Since the symbol ‘ 10^{13} ’ refers to the number 10^{13} and the utterance “ten to the thirteenth” verbalizes this symbol, the utterance likewise refers to the same number. That is, the utterance simply inherits its referent from the mathematical expression it verbalizes. Crucially, while the formal language of mathematics has its own syntactic and semantics rules, which determine that ‘ 10^{13} ’ refers to the number 10^{13} , these rules are external to natural language. We moreover hold that the conventions of math speak, that is, the conventions that govern the verbalization of terms in such formal languages, are likewise external to natural language grammar. If so, then the sequence *ten to the thirteenth* in (3) not only fails to derive its meaning from language-internal semantic composition, it also lacks any internal structure accessible to natural language grammar.

This analysis receives further empirical support from word order data. If a sequence of math speak lacks internal structure accessible to grammar, sub-extraction from it should be impossible. Thus, even though *to the thirteenth* in (3) resembles a regular prepositional phrase, and even though prepositional phrases in English routinely permit topicalization, such movement should not be possible in examples like (3). As demonstrated by the unacceptability of (24), this prediction is borne out.

- (24) *To the thirteenth, the distance from Earth to Proxima Centauri is four times ten kilometers.

Returning now to the examples in (6), we suggest that the existence of math speak opens up an avenue to their analysis. Given that *four times ten to the thirteenth* is a way of verbalizing mathematical symbols, it is at least conceivable that the same might be true for the *per*-phrases in (6). Concretely, it is possible that *thirteen grams per milliliter* and *thirty miles per hour* serve to verbalize expressions in the quantity calculus, viz. ‘ $13 \frac{\text{g}}{\text{mL}}$ ’ and ‘ $30 \frac{\text{mi}}{\text{hr}}$ ’, respectively. It would then be on those grounds that these sequences refer to the quotient quantities $13 \frac{\text{g}}{\text{mL}}$ and $30 \frac{\text{mi}}{\text{hr}}$. On this view, then, the meaning of certain *per*-phrases—those that refer to quotient quantities—would not be derived through composition within grammar. Therefore, there is no need to assume that *per* combines two simplex quantities into a complex one, as in (1), repeated below. The sentences in (6) thus no longer provide strong evidence for that entry. To make a proper case for (1), it would be necessary to rule out math speak as a viable alternative explanation for the data that the entry would serve to capture.⁷

⁷ Actually, while it seems plausible to us that the sequences *thirteen grams per milliliter* and *thirty miles per hour* can be math speak in their entirety, we do not wish to exclude that the numerals *thirteen* and *thirty* can alternatively combine with *grams per milliliter* and *miles per hour* compositionally, so that in (6), just the sequences *grams per milliliter* and *miles per hour* would be math speak. After all, given the common assumption that numerals compose with unit nominals within grammar in simpler sequences like *thirteen grams* or *thirty miles*, a parallel composition should be possible for *thirteen grams per milliliter* or *thirty miles per hour*. That said, as it will not affect our argument, we will for ease of exposition continue to sometimes include the numeral in the sequences that we propose to analyze as math speak.

$$(1) \quad \llbracket \text{per} \rrbracket = \lambda r. \lambda q. \frac{q}{r}$$

However, a math speak analysis is not merely possible, but is independently supported. We propose two kinds of evidence pointing in this direction, invoking substitution and restrictions on movement. Let us begin with substitution. Some phrases are unmistakable examples of math speak: their overt forms directly echo the symbols and configurations of the symbols used in the quantity calculus. The measure terms in (25) illustrate this pattern.

- (25) a. The density of that sample of liquid is thirteen gee over em el.
 b. The speed of that train is thirty em pee aitch.

The expression *thirteen gee over em el* verbalizes the symbol ' $13 \frac{\text{g}}{\text{mL}}$ ' whereas the phrase *thirty em pee aitch* verbalizes ' 30 mph ' which is equivalent to ' $30 \frac{\text{mi}}{\text{hr}}$ '. Within the quantity calculus, ' $13 \frac{\text{g}}{\text{mL}}$ ' and ' $30 \frac{\text{mi}}{\text{hr}}$ ' refer to the quotient quantities $13 \frac{\text{g}}{\text{mL}}$ and $30 \frac{\text{mi}}{\text{hr}}$. As instances of math speak, the corresponding verbalizations also refer to these complex quantities. Hence, if *thirteen grams per milliliter* and *thirty miles per hour* are also instances of math speak, the predicates in (25) are predicted to have the same denotations as those in (6), and thus the corresponding sentences are expected to be paraphrases of one another. This matches intuitions. The pairs of sentences differ in form but not in content.

Notably, although substitution is possible for sentences like those in (6), it is impossible when the terms *thirteen grams per milliliter* and *30 miles per hour* reference a simplex dimension like weight or distance, as in (8), repeated below. In those cases, substituting the measure phrases with *thirteen gee over em el* and *thirty em pee aitch* respectively, as in (26), yields non-sensical meanings that do not paraphrase the original statements in (8).

- (8) a. This sample of liquid weighs thirteen grams per milliliter.
 b. During its six-hour trip, the train (only) covered thirty miles per hour due to the frequent stops it had to make.
- (26) a. #This sample of liquid weighs thirteen gee over em el.
 b. #During its six-hour trip, the train (only) covered thirty em pee aitch due to the frequent stops it had to make.

This is expected, given the assumption that the terms *thirteen gee over em el* and *thirty em pee aitch* are math speak. Under such an assumption, these terms inherit their referents from the symbols ' $13 \frac{\text{g}}{\text{mL}}$ ' and ' 30 mph ', and as a result they refer to complex quotient quantities. As argued in Section 2, the terms *thirteen grams per milliliter* and *thirty miles per hour* in (8) are compositionally interpreted as simplex quantities. Thus, substitution should be prohibited.⁸

⁸The unacceptability of (26b) shows that it is not fully accurate to simply characterize the initialism *mph*, as an abbreviation for *miles per hour*, as dictionary entries often do. This characterization could suggest that (26b) can have a parse where it differs from its counterpart in (8b) in phonetic realization only, hence permits the same range of semantically interpreted syntactic structures. In turn, this would incorrectly predict that (26b) can be judged as felicitous, and equivalent to its counterpart in (8b). In contrast, if the term *mph* is unambiguously a math speak realization of ' $\frac{\text{mi}}{\text{h}}$ ', as we propose, it can only be used to refer to a measure of speed, not distance. The contrast between (8b) and (26b) is then correctly predicted.

The substitution facts demonstrate two points: a) math speak can be used to refer to complex quotient quantities and b) unambiguous cases of math speak have the same denotation as *per*-phrases that reference complex dimensions. Given this, the only additional and, seemingly, unexceptional assumption required under the math speak analysis is that ' $\frac{g}{mL}$ ' and ' $\frac{mi}{hr}$ ' also permit the verbalizations *grams per milliliter* and *miles per hour*.

Restrictions on syntactic movement add further support for this analysis. Recall that sub-sequences of math speak are syntactically invisible to movement operations within natural language grammar. This was already shown in (24), with an attempted sub-extraction of *to the thirteenth* from *four times ten to the thirteenth*. Now, if the sequences *grams per milliliter* and *miles per hour* are instances of math speak in the sentences in (6), then sub-extraction should likewise be impossible. In particular, even though the *per milliliter* and *per hour* look like prepositional phrases, which are generally mobile in English grammar, *per milliliter* and *per hour* are expected to be immobile when they are part of a measure term that refers to a complex quotient quantity. The data in (27) and (28) show that this prediction is correct.

- (27) a. #The density of that sample of mercury is, per milliliter, thirteen grams.
b. #The density of that sample of mercury, per milliliter, is thirteen grams.
c. #Per milliliter, the density of that sample of mercury is thirteen grams.
- (28) a. #The speed that this train is travelling at is, per hour, thirty miles.
b. #The speed that this train is travelling at, per hour, is thirty miles.
c. #Per hour, the speed that this train is travelling at is thirty miles.

Intuitively, these sentences are awkward because they attempt to equate a measurement of weight and distance to a degree of *density* and *speed*, respectively. This is precisely the intuition that is expected under a math speak analysis. In each case, sub-extraction requires a compositional interpretation, which, in turn excludes the possibility that the measure term could refer to a quotient quantity. In contrast, such sub-extraction *is* available, as predicted, for the fully compositional data in (8), as shown by the paraphrases in (29) and (30).

- (29) a. This sample of liquid weighs, per milliliter, thirteen grams.
b. This sample of liquid, per milliliter, weighs thirteen grams.
c. Per milliliter, this sample of liquid weighs thirteen grams.
- (30) a. During its six-hour trip, the train covered, per hour, (only) thirty miles due to the frequent stops it had to make.
b. During its six-hour trip, the train, per hour, covered (only) thirty miles due to the frequent stops it had to make.
c. Per hour, during its six-hour trip, the train covered (only) thirty miles due to the frequent stops it had to make.

The freedom of movement in (29) and (30) contrasts sharply with the awkwardness of (27) and (28). If all *per*-phrases were assumed to have structures given by natural language grammar, it would seem hard to explain why these superficially identical *per*-phrases behave so differently in the two sets of data. In contrast, the hypothesis

that one measure expression has internal structure while the other is an instance of math speak naturally captures the difference.

In sum, we have in this section seen evidence that not all *per*-phrases receive a compositional interpretation within natural language grammar. Certain *per*-phrases, like *pounds per square inch* can be non-compositional by virtue of being idioms. More broadly, we have argued that even productive *per*-phrases can have meanings that are not composed in grammar, but arise from language external conventions. We have made this case for *per*-phrases that saturate predicates of quotient quantities, like density or speed. We argued that such expressions are instances of math speak and that their meanings are inherited from the language external terms that they verbalize.

If our arguments are accepted, then it is not necessary to assume that *per* can express quantity division as in (1), mapping a pair of quantities to quotient quantities. In fact, our account of how sub-extraction from *per*-phrases is constrained is incompatible with such an interpretation. This in turn raises the further question of *why* a lexical entry like (1) would not be available. The semantic type that (1) assigns to *per*, $\langle d, \langle d, d \rangle \rangle$, seems unexceptional. Therefore, if there is a broader generalization behind the unavailability of (1), it would presumably have to do with the operation of quantity division that *per* performs according to (1). This leads us to the No Division Hypothesis stated in (5) in Section 1, repeated here.

- (5) **No Division Hypothesis:** Quantity division is not available as an operation for semantic composition.

This hypothesis goes beyond the rejection of (1) in that it makes the stronger proposal that *no* lexical item or semantic rule could reference quantity division. It implies that any compositional interpretation that relies on implementing division lies outside the domain of natural language semantics. We are not in a position to fully motivate a deeper explanation for why this might be so. We note, however, that while some mathematical operations (e.g., addition and multiplication) are rooted in intuitive practices of grouping and scaling, quantity division represents a more abstract operation, which notably was explicated much later in the history of mathematics (see, e.g., de Boer 1995). Given this, we suggest, it would actually be surprising if division were engrained in the grammar of natural language.

In the remainder of this paper, we focus on the central assumption exploited in this section, viz. that natural language sentences can be infiltrated by expressions whose meanings are determined by language external conventions. We point out that this possible interplay between internal linguistic structure and external conventions is not unique to *per*-phrases. Similar dynamics can be observed in so-called mixed quotation, where language conveys semantic content (e.g., what is signified) while simultaneously describing the characteristics of a symbolic system being used to relay that content. By turning to mixed quotation, we hope to further explore how language integrates external references, shedding light on the broader interpretive mechanisms at play in expressions like *per*-phrases and beyond.

3.2 Mixed quotation and formal languages

As argued in the last section, natural language can be infiltrated by expressions that instantiate math speak—expressions that verbalize terms in technical symbolic systems. Note now that such expressions serve two purposes: they describe the symbols being used, and at the same time they denote the values related to those symbols. We propose that the non-compositional interpretations of *per*-phrases in such contexts bear intriguing parallels to the phenomenon of “mixed quotation.” Here we will not attempt to engage with the theory of mixed quotation in depth. Instead, we merely hope to make it plausible that any successful account of mixed quotation could naturally be extended to explain the non-compositional interpretations of *per*-phrases, unifying these two phenomena under a common umbrella.

As discussed in Davidson (1979), speakers will often use quotations to serve two functions: to signal that something was “said” and to denote a specific meaning. For example, consider the sentence in (31) from Davidson’s original paper.

- (31) Quine says that quotation “...has a certain anomalous feature.”

The quoted content in (31) not only conveys what was said by Quine, but also uses the semantic content of the quoted expression to form a proposition. This “double duty”, as Davidson (1979) called it, can be brought to the fore by substituting the verb *says* with *believes*.

- (32) Quine believes that quotation “...has a certain anomalous feature.”

The sentence in (32) both states the content of Quine’s belief while simultaneously conveying the exact words said by Quine, a characteristic which disappears if the quotes are removed. Due to its dual nature, such instances of quotation have been labelled as *mixed quotations*, that is, expressions that mix *mention* and *use*.⁹

Critical to the issue at hand, mixed quotations are not limited to expressions within a speaker’s grammar. For example, a speaker can use the quoted pronunciation “[eɪ]pricots” to denote apricots, as shown in (33a), even if this pronunciation is not part of the speaker’s grammar. Similarly, a speaker can use “eckullectic” to mean the same thing as *eclectic*, as in (33b), even if the speaker is not familiar with all the nuances of the idiolect spoken by George W. Bush.

- (33) a. When in Santa Cruz, Peter orders “[eɪ]pricots” at the local market. When in Amherst, Peter orders “[æ]pricots” at the local market. (example from Potts 2007)
b. Bush is proud of his “eckullectic” reading list. (example from Shan 2010)

⁹Partee (1973) used the term “mixed direct and indirect quotation” to label similar examples, and perhaps this is the origin of the turn of phrase. However, it is important to note that Partee questioned whether such instances of quotation were a part of natural language. Within the same paragraph where she introduces the term, she also states her hunch that “such sentences do not occur in ordinary spoken language” (p. 411). Partee’s comments are very similar to those made by Morzycki (2017) concerning the possible existence of *semantic viruses*, extra-grammatical rules that affect language use without being a part of the core grammar. Whether mixed quotation has a grammatical or extra-grammatical explanation, what is critical for the issue at hand is that at least some uses of a *per* measure-phrase are arguably not derived compositionally from the lexical entry for *per*.

Not only can quoted expressions vary in terms of phonological representations within a speaker’s grammar, they can also violate a speaker’s morpho-syntactic rule system. For example, a speaker for whom the expressions “bestest friend” and “on accident” are ungrammatical (e.g., your average adult Canadian English speaker) can still use those expressions in quotation to denote the same thing as *best friend* and *by accident* respectively, as shown in (34a) and (34b). (*Bestest* is a common expression used by children, whereas *on accident* is a common expression within many dialects of American English.) Similarly, a speaker who is not fluent in French can still use the expression “vieille femme hystérique” (old hysterical woman) in quotation, as shown in (34c).

- (34) a. George says Tony is his “bestest friend”. (example from [Geurts and Maier 2005](#))
 b. Meagan said she took out the trash “on accident”. (example from [Cappelen et al. 2020](#))
 c. A doctor tells him [Gustave Flaubert] he is like a “vieille femme hystérique”; he agrees. (example from TLS online, 18 December 1998, as cited in [De Brabanter 2010](#))

This type of quotation can even be used when a speaker is unsure about the meaning of a quoted phrase. For instance, consider (35) in a context where the speaker is an experienced but naive naturalist who is trying to learn the technical order, family and genus names assigned in biology.

- (35) I took a class with Professor Wilson. Apparently, there are many different species of “Lagomorpha Leporidae Sylvilagus” in Canada. I wonder if I have ever seen one?

The propositions conveyed by the last two sentences in (35) have truth conditions that depend on whether there are different species of cottontail rabbits living in Canada, although the speaker doesn’t need to know how the biological sciences define “Lagomorpha Leporidae Sylvilagus” in order to use these Latin terms in this way. Rather, by quoting, the speaker is relying on the expertise of Professor Wilson, or perhaps on the biological discipline as a whole, to establish the sentences’ truth conditions.

A different, but related, feature of quotation in everyday usage is that it need not involve literal repetitions of something previously said (in contrast to what would be required in, say, academia writing). As pointed out by [Sams \(2010\)](#), people often quote hypothetical and future speech. Sams provides a wealth of examples, including the sentence in (36) taken from a recorded conversation between two friends.

- (36) You’re going to be telling me that on my death bed your mom made the “best grilled cheese.”

In spoken speech, quoted phrases are frequently marked by a change in voice, sometimes imitating some aspect of the person being quoted (see, [Sams, 2010](#)). In this case, “best grilled cheese” was uttered with a breathy intonation distinct from the rest of the sentence. Of critical importance to the issue at hand, the quoted phrase in (36)

does not repeat anything that was said, but rather it quotes an ironic and hypothetical statement that might occur in the future.

Taking stock, since speakers can (mix-)quote (i) grammars that they do not fully know, (ii) scientific uses of language, and (iii) hypothetical speech, it seems reasonable to suspect that speakers should be able to use math speak to quote pseudo-scientific technical speech that relates to mathematical and scientific notation. Perhaps this is what speakers are doing when they use *thirty miles per hour* to refer to a degree of speed or *two grams per milliliter* to refer to a degree of density. They are “quoting” vocalizations of technical terms within an artificial grammar.

Finally, viewing math speak as an instance of mixed quotation is also consistent with the restriction on sub-extraction observed in the last section. Based on the *wh*-movement example in (37), Maier (2014) suggests that mixed quotations do not permit sub-extraction. Maier imagines a speaker who did not properly hear the end of the mixed quotation example in (31) above, and claims that it would be deviant for that speaker to respond with (37). If Maier’s judgment is correct, then this illustrates resistance to sub-extraction, on the assumption that the verb phrase in this response must be parsed as a mixed quotation.

(37) ??What did Quine say quotation “has”?

This restriction can be credited to the assumption that mixed quotations are not structurally integrated in the syntactic context that surrounds them, just as we assumed for math speak. Hence, viewing math speak as mixed quotation preserves our proposed interpretation of the observed restriction on movement.¹⁰

How in general speakers convey meanings through mixed quotations is a subject that deserves a thorough investigation, especially given that speakers do not need to know the grammar they are quoting in order to successfully communicate. However, we will not provide such an investigation here. What is important with respect to our focus in this paper is that people regularly use utterances that are related to symbols in potentially unknown or partially-known grammars, relying on external conventions rather than their own linguistic grammar to interpret and convey meaning. It seems feasible that this ability could account for the interpretation of *per*-phrases when they are used to refer to quotient quantities that express measurements in complex dimensions like speed or density.

4 Conclusion

This paper has critically examined the role of *per* in natural language, particularly the claim that *per* introduces quotient quantities through a division operation. Our analysis has revealed that this interpretation can be questioned: it fails to account for

¹⁰ At the same time, building on Shan (2010), Maier suggests that a process of *unquotation* can create the appearance of *wh*-movement from a mixed quotation in data like (i). Under Maier’s analysis, then, given that movement from terms in math speak, including *per*-phrases, seems to be invariably excluded, we would have to assume that unquotation, however explicated, is systematically unavailable within such terms.

(i) Who did Mary say that she would “never misundereestimate ever again”?

key data where *per* interacts with measure verbs like *weigh* and *cover*, and it imposes unnecessary complexity on the compositional semantics of *per*-phrases.

In response, we have proposed an alternative account in which *per* does not rely on quantity division but instead interacts with pure numbers and multiplication. This reanalysis avoids the complications associated with quotient quantities while providing a straightforward and empirically grounded explanation for the behavior of *per* in phrases like *thirteen grams per milliliter*.

Our investigation also highlights a broader interpretive duality with *per*-phrases: while some uses align with compositional interpretations rooted in the syntactic and semantic structure of natural language, others depend on extra-grammatical conventions, akin to mixed quotation. Specifically, we argued that phrases like *thirteen grams per milliliter* and *thirty miles per hour* often function as verbalizations of technical notation. These instances bypass semantic composition and instead reflect an external alignment with systems like the quantity calculus.

Although our suggestions about math speak echo remarks made in [Bylinina and Nouwen \(2020\)](#) and [Morzycki \(2017\)](#) (cf. footnote 2), our paper may be the first to propose that natural language can use quotations from technical grammars to denote the very entities and quantities those grammars were designed to describe. Given that speakers can verbalize mathematical notation—a routine practice for teachers in these fields—it seems only natural that verbalizations of such notation could be quoted in the same way as phrases involved in canonical instances of mixed quotation. This capacity underscores the adaptability of natural language in incorporating external systems of representation.

While our discussion of this phenomenon in this paper is necessarily limited, we believe that this extension of mixed quotation holds significant potential for advancing the analysis of the language of numbers and quantities. Specifically, it raises the possibility that certain phrases in natural language may lack compositional meaning within a speaker’s grammar and instead function as quoted chunks whose compositional interpretation exists only within a technical grammar. To support this claim, we have outlined two diagnostics for identifying when an expression is being used as a mixed quote: the ability to substitute arbitrary notational descriptions, and sensitivity to syntactic movement. These diagnostics, though not exhaustive, provide a framework for further investigation. They are expected to aid the study of further grammatical phenomena that elucidate the general nature of mixed quotation. We hope to explore these avenues in future work.

To explain why languages would prohibit lexical entries such as (1), we posited the *No Division Hypothesis*, which suggests that natural language semantics does not include a compositional operation akin to division. The findings presented in this paper lend support to this hypothesis. While the quantity calculus relies on division to derive quotient dimensions like density and speed, our analysis demonstrates that natural language achieves reference to these dimensions through alternative mechanisms. The *No Division Hypothesis* would explain why natural language appears to rely on pseudo-technical expressions, mixed quotation, and other external conventions to gain access to the derived scales of the quantity calculus.

A natural direction for future research would explore how quantities are expressed cross-linguistically. The *No Division Hypothesis* makes the strong prediction that quantity division is not available as an operation in the grammar of *any* language. At the same time, languages may well differ in terms of whether they permit reference to quotient quantities. We predict that such reference is available in principle for languages that, like English, permit mixed quotation of technical notation, but will be unavailable otherwise. Finally, the availability of compositional *per*-phrases in any given language is expected to be a matter of lexical inventory, and there may well be languages that lack such phrases. These predictions invite systematic cross-linguistic testing.

Competing Interests

This research was partially funded by the Social Sciences and Humanities Research Council (SSHRC) of Canada (Grant numbers 435-2023-0699 and 435-2023-0218). One of the authors is an associate editor at *Linguistics and Philosophy*.

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